

Ex. No. : 01

Date:

Register No.: 231701017

Name: HARISH BABU R

A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE, BIVARIATE AND MULTIVARIATE REGRESION

Aim:

Develop a Python program to implement **univariate, bivariate, and multivariate regression** models and analyze the relationship between dependent and independent variables.

Requirements:

Jupyter Notebook with Python, Diabetes.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from google.colab import files
```

```
uploaded = files.upload()
```

```
df = pd.read_csv('Iris.csv')
df = df.drop(columns=['Id'])
```

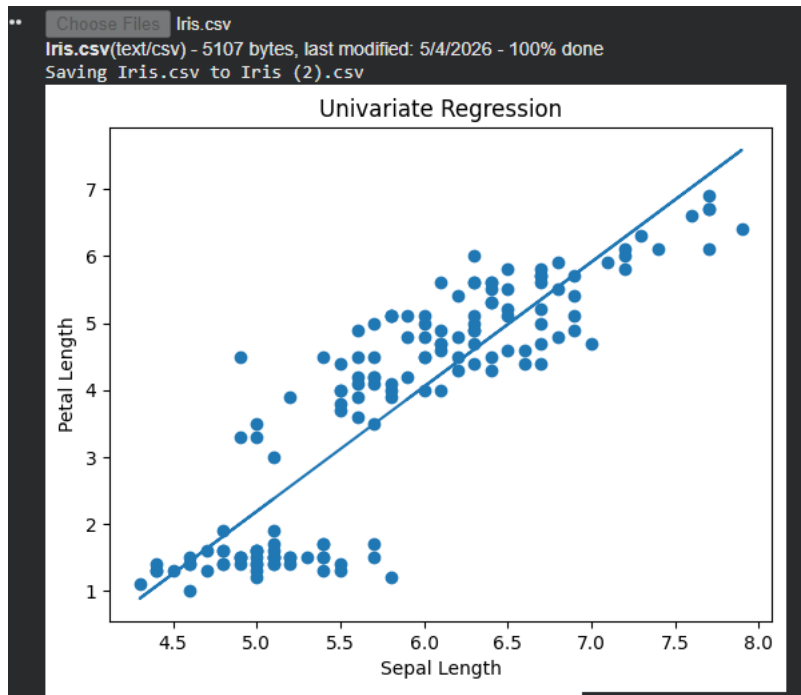
```
Y = df['PetalLengthCm'].values
```

```
def linear_regression(X, Y):
    X = np.c_[np.ones(X.shape[0]), X]
    return np.linalg.inv(X.T @ X) @ X.T @ Y
```

```
# 1. Univariate Regression
X1 = df[['SepalLengthCm']].values
theta1 = linear_regression(X1, Y)
```

```
plt.figure()
```

```
plt.scatter(X1, Y)
plt.plot(X1, theta1[0] + theta1[1]*X1)
plt.title("Univariate Regression")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()
```



```
# 2. Bivariate Regression
X2 = df[['SepalLengthCm',
'SepalWidthCm']].values
theta2 = linear_regression(X2, Y)

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(X2[:,0], X2[:,1], Y)

xx, yy = np.meshgrid(
    np.linspace(X2[:,0].min(), X2[:,0].max(),
10),
    np.linspace(X2[:,1].min(), X2[:,1].max(), 10)
)

zz = theta2[0] + theta2[1]*xx + theta2[2]*yy

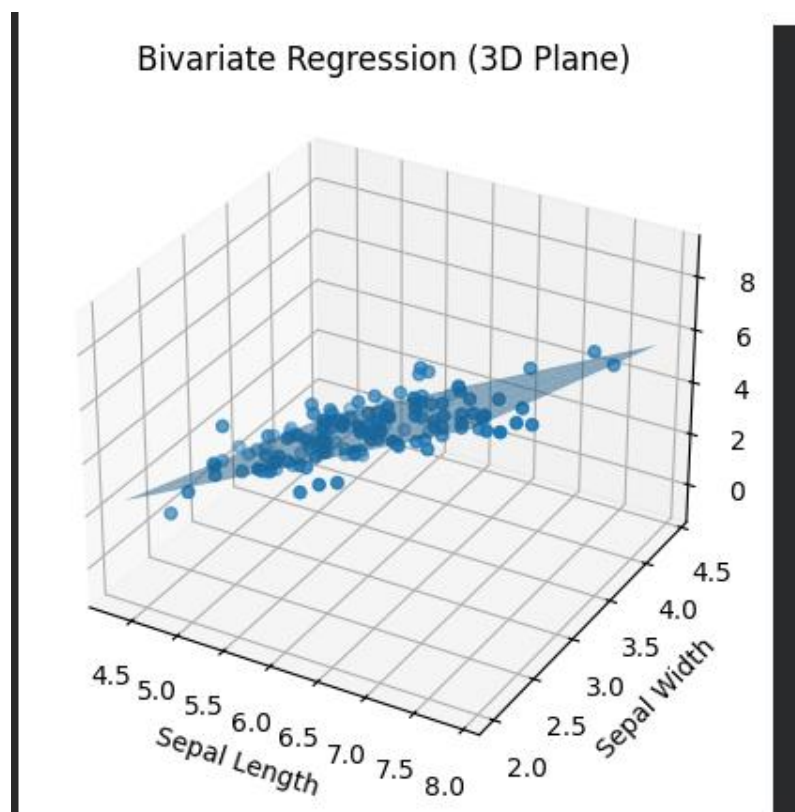
ax.plot_surface(xx, yy, zz, alpha=0.5)
```

```

ax.set_title("Bivariate Regression (3D
Plane)")
ax.set_xlabel("Sepal Length")
ax.set_ylabel("Sepal Width")
ax.set_zlabel("Petal Length")

plt.show()

```



3. Multivariate Regression

```

X3 = df[['SepalLengthCm', 'SepalWidthCm',
'SepalWidthCm', 'PetalWidthCm']].values
theta3 = linear_regression(X3, Y)

```

```

X3_bias = np.c_[np.ones(X3.shape[0]), X3]
Y_pred = X3_bias @ theta3

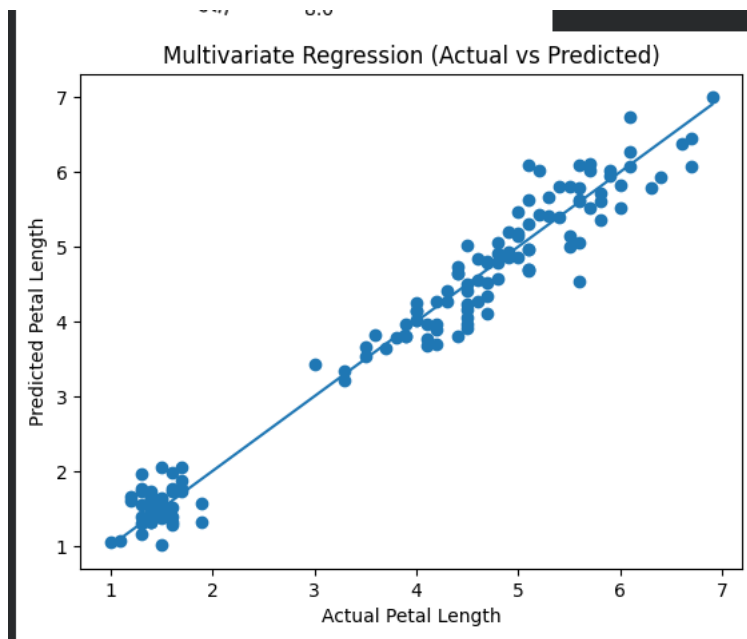
```

```

plt.figure()
plt.scatter(Y, Y_pred)
plt.plot([min(Y), max(Y)], [min(Y), max(Y)])
plt.title("Multivariate Regression (Actual vs
Predicted)")
plt.xlabel("Actual Petal Length")
plt.ylabel("Predicted Petal Length")

```

```
plt.show()plt.title("Univariate Analysis -  
BMI") plt.show()
```



Result:

the Python program for univariate, bivariate, and multivariate regression was successfully implemented using the Diabetes.csv dataset in Jupyter Notebook, and the regression models were able to analyze and predict relationships between variables accurately.